

The Open Source Way to EU Digital Sovereignty & Competitiveness

A Strategic Roadmap for Action

*Based on the July 2025 Report by the European Alliance
for Industrial Data, Edge and Cloud*

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EUROPEAN ALLIANCE
FOR INDUSTRIAL DATA,
EDGE AND CLOUD

THEMATIC ROADMAP





Who am I?

- Linux user since 1991
- Open Source advocate since 1998
- Open Source entrepreneur since 2000 (Nuxeo -> Abilian)
- Co-chair of CNLL, APELL
- Co-founder of the EuroStack Initiative Foundation
- Member of the European Alliance for Industrial Data, Edge and Cloud, and co-author of the report

The Strategic Context: A Digital Crossroads

Europe's digital infrastructure is increasingly built on technologies developed and controlled outside our borders. What was considered by many a merely technical issue has now become a major strategic vulnerability.

This dependency creates significant risks for our:

- **Economic Autonomy:** Vulnerability to non-EU market dominance (€265 billion / year just for B2B cloud software and services from EU->US) .
- **Data Security:** Exposure to extraterritorial laws like the FISA and CLOUD Act, exposing personal information (cf. GDPR) and strategic / corporate information and secrets.
- **Strategic Independence:** A diminished ability to innovate according to our own values.

TECHNOLOGICAL DEPENDENCE ON
AMERICAN SOFTWARE AND CLOUD
SERVICES: AN ASSESSMENT OF THE
ECONOMIC CONSEQUENCES
IN EUROPE

Economic study
April 2025

ASTERès
études, recherche & conseil économique

Defining Our Goal: Effective Digital Sovereignty

To be truly sovereign, we need more than just control over data. The report starts from a simple and clear formula:

Digital Sovereignty = Data Sovereignty + Technological Autonomy

This means we must control the entire stack—the underlying hardware and software that manages our data. This technological independence is the key to effective sovereignty.

The Enabler: Why Open Source is the Cornerstone

Initially (1998-2010) considered mostly as a cheaper alternative, Open Source has become for the last 15 years a fundamental pillar of the technological autonomy Europe seeks.

- **Autonomy & Reversibility**

It is the only real escape from vendor lock-in. Full control over the code guarantees the freedom to innovate independently.

- **Collaboration & Innovation**

It allows us to pool resources to build cutting-edge technologies that no single entity could develop alone.

- **Transparency & Security**

The ability to audit code allows for collective verification, vulnerability fixing, and ensures no hidden backdoors.

From Diagnosis to a Blueprint for Action

To build a credible plan, we needed a deep diagnosis. Our analysis of Europe's digital landscape revealed a fundamental disconnect between our ambition for sovereignty and the reality of our ecosystem.

This diagnosis identified **5 critical "gaps"**. From these gaps emerged **70 concrete proposals**, each designed to address a specific weakness and build a stronger whole.

Gap 1: Standards & Interoperability

The Challenge of "Open Washing"

The Problem: Our ecosystem is fragmented by "standards" that are not genuinely designed to ensure interoperability and reversibility, promoted by non-EU players to maintain dominance. This creates vendor lock-in and stifles innovation.

The Consequence: A lack of genuine, enforceable interoperability that keeps Europe dependent and divided. Cf. the battle for EIFv2 (2009).

Gap 2: Financial Viability

The Peril of Instability

The Problem: Many critical European Open Source projects and businesses rely on sporadic funding, time-limited contracts and volunteer work. They lack resources for security audits, long-term maintenance, whole product development and professional support.

The Consequence: Key infrastructure components are unstable, uncompetitive, and at risk of being abandoned or acquired by non-EU interests.

Gap 3: Market Visibility & Adoption

The Fight for Recognition

The Problem: European Open Source solutions are often invisible, drowned out by the dominant marketing narratives of non-EU giants.

The Consequence: Public and private sectors alike default to non-sovereign solutions, not because they are better, but because they are better known.

Gap 4: Talent & Skills Shortage

The Human Capital Deficit

The Problem: The lack of expertise in sovereign Open Source technologies slows down innovation and implementation.

The Consequence: A persistent skills gap makes us reliant on foreign expertise, directly undermining our goal of technological autonomy.

Gap 5: Governance & Influence

The Imbalance of Power

The Problem: Europe contributes a vast amount of code to global Open Source projects (more than the US!) but exercises too little control over their strategic direction, which is often decided in foundations based outside the EU.

The Consequence: Key technological dependencies remain, as the roadmaps of critical software are not aligned with European strategic interests.

The Blueprint: A 5-Pillar Strategy to Bridge the Gaps

Our 70 proposals are organized into a clear, actionable strategy built on five pillars. This is our roadmap for a sovereign digital future for Europe.

- 1. Technological Development**
- 2. Skills Development**
- 3. Public Procurement**
- 4. Growth and Investment**
- 5. Governance**

(Origin, in large part: previous proposals and reports from APELL, CNLL, OSBA, parliamentary reports, European projects, etc.)

Pillar 1: Technological Development

Objective: To build a resilient technical foundation based on truly open standards and well-supported European projects.

Emblematic Actions:

1. **Create a European Open Source Sovereignty Fund (EOSSF)**

To provide stable, long-term funding for the core infrastructure we all depend on.

2. **Define and Enforce TRULY Open Standards**

Develop European-governed specifications to fight "open washing" and ensure genuine interoperability.

3. **Launch Large-Scale Demonstration Projects**

Co-invest in pilot projects that prove the technical and economic viability of sovereign stacks in key sectors.

4. **Develop and Promote EU Reference Implementations**

To showcase excellence and provide practical blueprints for adoption.

Pillar 2: Skills Development

Objective: To bridge the talent gap by fostering a skilled workforce proficient in European Open Source technologies.

Emblematic Actions:

1. Establish European "Centres of Excellence"

Support universities in becoming hubs for advanced research, training, and innovation in sovereign open source technologies.

2. Launch Pan-European Training Programs and Certifications

Create industry-recognized certifications for European Open Source skills, making them a gold standard for professionals and employers.

3. Integrate Open Source into STEM Curricula

Embed open source principles and digital sovereignty concepts into education from secondary school onwards.

4. Invest in Reskilling and Upskilling Initiatives

To adapt the existing workforce to meet new technological demands.

Pillar 3: Public Procurement

Objective: To leverage public spending as a powerful catalyst for the adoption of European Open Source solutions.

Emblematic Actions:

1. **Mandate "Public Money, Public Code, European Preference"**

Adopt this powerful principle to ensure that taxpayer-funded software originates from, and benefits, the European ecosystem.

2. **Define Clear Criteria for "European Open Source"**

Establish a robust definition to prevent misuse and ensure public funds support genuinely sovereign projects.

3. **Support Evaluation and Guidance for EU Open Source Solutions**

To empower public bodies to make informed, sovereign choices.

4. **Leverage Pre-Commercial Procurement for EU Open Source Development**

To co-develop innovative solutions tailored to Europe's public sector needs.

A Cornerstone Proposal: Defining "European Open Source"

To make policies like "European Preference" effective, we must first have a clear, robust definition to prevent "open washing" and "sovereign washing".

Proposed Criteria for a "European Open Source" Solution:

- **Origin of Development**
A significant portion of R&D and development is conducted in Europe.
- **Governance**
The project's governance is based in Europe and aligned with EU values.
- **Community**
A substantial part of the user and contributor base is European.
- **Legal Framework**
Uses OSI-approved licenses compatible with EU law.
- **Data Handling**
Prioritizes data processing and residency within the EU, in compliance with EU laws and directives (like GDPR, but non only).
- **Ecosystem Contribution**
The maintaining organization actively contributes back to EU ecosystem.

From Definition to Action: A New Procurement Principle

Building on this definition, the report proposes a powerful new policy for public spending:

Public Money, Public Code, Open Source First, European Preference

This extends the well-known "Public Money, Public Code" initiative. The "**European Preference**" clause ensures that taxpayer-funded software not only remains open, but also **originates from and benefits the European Open Source ecosystem**.

It transforms public procurement from a simple purchase into a strategic investment in our own digital future.

Pillar 4: Growth and Investment

Objective: To build a robust and sustainable funding ecosystem for the growth of European Open Source.

Emblematic Actions:

1. Build a Strong "European Open Source" Brand

Develop a global brand that signifies quality, security, and alignment with European values, helping our solutions stand out.

2. Launch a European OS Investment Platform (EOSIP)

Create a one-stop-shop portal to connect European Open Source projects with grants, venture capital, and other funding opportunities.

3. Establish European Open Source Accelerators

Launch dedicated incubators to help promising projects scale up and compete globally.

4. Support Public-Private Partnerships to Drive Innovation

To pool resources and expertise for high-impact projects.

Pillar 5: Governance

Objective: To ensure the long-term security, sustainability, and European control of critical Open Source projects.

Emblematic Actions:

1. Form a European Open Source Advisory Council

Establish a high-level body of experts to guide funding, promote best practices, and ensure critical projects remain under European stewardship.

2. Align with Existing Compliance and Data Sovereignty Standards

Ex.: Offer dedicated resources to help European projects achieve CRA certification, turning compliance into a mark of quality and trust.

3. Promote Active Participation in Governance

Incentivize and support the active involvement of European experts in the decision-making bodies of key international foundations.

4. Ensure Long-Term Sustainability for Critical European Projects

To provide stable stewardship for foundational technologies.

The Added Value: Transformative Impact on Key Sectors

Implementing this roadmap will unlock concrete benefits across our society:

- **Public Administration**

Enhances security, transparency, and accountability while reducing dependency and lowering costs.

- **Manufacturing (Industry 4.0)**

Enables predictive maintenance, improves interoperability on the factory floor, and fosters innovation.

- **Healthcare**

Ensures patient data security, enables system interoperability, and accelerates medical research.

- **Energy**

Supports a greener future by optimizing grids, integrating renewables, and reducing data center energy use.

The Business Case for European Companies

1. Growth & Profitability

- **Use Trust as a Market Differentiator:** Win premium business with a "GDPR-native" guarantee.
- **Unlock New Market Opportunities:** Align with EU policy to win public sector contracts.

2. Operational Efficiency & Cost Reduction

- **Avoid Vendor Lock-In:** Control Total Cost of Ownership (TCO) with open standards.
- **Reduce the Compliance Burden:** Eliminate the cost of managing international data transfers.

3. Strategic Resilience & Risk Mitigation

- **Protect Intellectual Property:** Insulate your business from extra-territorial laws like FISA/CLOUD Act.
- **Secure the Digital Supply Chain:** Reduce geopolitical risk and ensure business continuity.

A Vision for Europe: A Virtuous Cycle of Sovereignty

Implementing this roadmap will create a self-reinforcing cycle:

- **Sovereign Infrastructure** will give us control over our digital destiny.
- **A Thriving Ecosystem** will empower SMEs, create high-skilled jobs, and drive innovation.
- **Reduced Dependency** will make our economy more resilient and secure.
- **Technology Aligned with Our Values** will ensure that Europe's digital future is open, transparent, and human-centric.

Conclusion: A Call to Action

Achieving European digital sovereignty requires the collective support and action of our entire ecosystem—from policymakers and industry leaders to developers and educators. Additional initiatives include:

- The "Cloud Sovereignty Framework" from the Commission
- Various public procurement initiatives (EU and MS)
- The EU-STF (European Sovereign Tech Fund) initiative
- APELL's "Digital Declaration of Independence" (contact me)
- Etc.

Let's build our digital future, the Open Source way.

Get the full report: <https://ec.europa.eu/newsroom/dae/redirection/document/117980>